

ECSE 2500

Lee '16

Mar. 23th

Topic: ① PDF/CDF of a function of continuous RV

② Markov / Chebyshev inequality

③ Introduce PMF/PDF of two RVs

□ PDF/CDF of a function of RVs

Talk about $Y = g(X) = X^2$. Now we are given CDF of X as $F_X(x)$, for all x . We want to find CDF/PDF of Y ?

Answer: $F_Y(y) = P(Y \leq y)$

$$= P(X^2 \leq y)$$

$$= P(-\sqrt{y} \leq X \leq \sqrt{y})$$

$$= P(X \leq \sqrt{y}) - P(X \leq -\sqrt{y})$$

$$= F_X(\sqrt{y}) - F_X(-\sqrt{y}), y \geq 0$$

Once we have the CDF of Y , we can derive PDF of y via derivative

$$\begin{aligned} f_Y(y) &= \frac{\partial F_Y(y)}{\partial y} \\ &= \frac{\partial F_X(\sqrt{y})}{\partial \sqrt{y}} \cdot \frac{\partial \sqrt{y}}{\partial y} + \frac{\partial F_X(-\sqrt{y})}{\partial (-\sqrt{y})} \cdot \frac{\partial (-\sqrt{y})}{\partial y} \\ &= f_X(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} + f_X(-\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} \\ (*) \quad &= \frac{1}{2\sqrt{y}} (f_X(\sqrt{y}) + f_X(-\sqrt{y})) \end{aligned}$$

Example: Assume X is a Gaussian RV with mean $\mu=0$, variance $\sigma^2=1$.
Let $Y = X^2$. What is PDF $f_Y(y)$?

Answer: PDF of X

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

Plugging the Gaussian PDF into (*)

$$\begin{aligned} f_Y(y) &= \frac{1}{2\sqrt{y}} \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{y}{2}} + \frac{1}{\sqrt{2\pi}} e^{-\frac{y}{2}} \right) \\ &= \frac{1}{\sqrt{y}} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{y}{2}}, \quad y \geq 0 \end{aligned}$$

Remark: This is called a chi-square (χ^2) random variable with one degree of freedom.

- If $Y = g(X) = |X|$, then what is PDF/CDF of Y ?

By definition of CDF,

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(|X| \leq y) \\ &= P(-y \leq X \leq y) \\ &= P(X \leq y) - P(X \leq -y) \\ &= F_X(y) - F_X(-y), \quad y \geq 0 \end{aligned}$$

With CDF, we can derive PDF of y as

$$f_Y(y) = \frac{dF_Y(y)}{dy} = f_X(y) + f_X(-y)$$

- We know CDF and PDF of X as

$$F_X(x) \text{ and } f_X(x). \text{ If } Y = g(X) = \begin{cases} X, & \text{if } X \geq 0 \\ 0, & \text{o.w.} \end{cases}$$

What is the CDF / PDF of Y ? $= (X)^+$

$$F_Y(y) = P(Y \leq y)$$

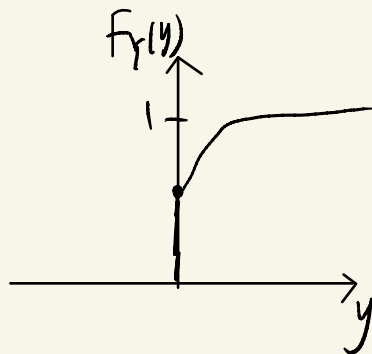
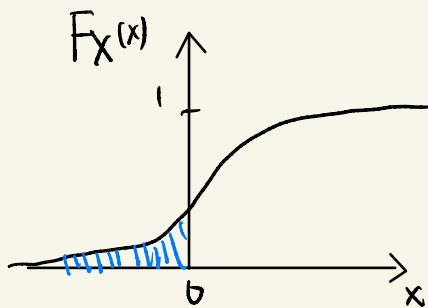
$$= \begin{cases} P(X \leq y) & \text{if } y \geq 0 \\ 0 & , \text{ otherwise} \end{cases}$$

$$= \begin{cases} F_X(y) & , \text{ if } y \geq 0 \\ 0 & , \text{ o.w.} \end{cases}$$

$$f_Y(y) = \begin{cases} f_X(y) & , \text{ if } y \geq 0 \\ 0 & , \text{ o.w.} \end{cases}$$

$$f_Y(y) = F_X(0) \delta(y) + f_X(y) u(y)$$

Visualization



$$f_Y(y) = \underbrace{\quad}_{F_Y(0)} + \underbrace{\quad}_{\text{PDF of } X}$$

Example: X is a Gaussian RV with mean $\mu=2$, and variance $\sigma^2=4$. If $Y=(X)^+$, compute PDF of Y $f_Y(y)$?

Using the conclusion in the above derivation

$$\begin{aligned} f_Y(y) &= F_X(0) \delta(y) + f_X(y) u(y) \\ &= \phi\left(\frac{0-2}{2}\right) \delta(y) + \frac{1}{\sqrt{2\pi} \cdot 2} e^{-\frac{(y-2)^2}{4}} u(y) \end{aligned}$$

Summary

In general, if $Y = g(X)$ and the derivative $g'(x)$ exists, the general form of PDF of Y can be represented as follows:

Let y be a given value of Y and let $\{x_1, x_2, \dots, x_k\}$ denote the set of values of x that $g(x) = y$, then we can write

$$f_Y(y) = \sum_{i=1}^k f_X(x_i) \cdot \frac{1}{|g'(x_i)|}$$

Apply to $g(X) = X^2$? *Verify it offline.*

□ Some inequalities/bound about CDF/PDF.

Say we only know RV X is non-negative with mean μ . Can we say something about $P(X)$?

Markov inequality

$$P(X \geq a) \leq \frac{E[X]}{a}$$

↑ ↑
non-negative

Proof :

By definition of mean of X ,

$$\mu = E[X] = \int_0^{\infty} x f_X(x) dx$$

$$= \int_0^a x f_X(x) dx + \int_a^{\infty} x f_X(x) dx$$

$$\geq \int_a^{\infty} x f_X(x) dx$$

$$\geq a \int_a^{\infty} f_X(x) dx = a P(X \geq a)$$

$$\Leftrightarrow P(X \geq a) \leq \frac{E[X]}{a}$$

Markov inequality gives us an upper bound on the probability of RV X from its mean $E[X]$.

Example: (Application of Markov inequality)

Printer gets crashed if it gets more than 1000 requests/sec. We measure the average workload of the printer as 50 requests/sec. What is the estimate of $P(\text{Printer gets crashed})$?

Answer: $P(\text{Crash}) = P(X \geq 1000)$

Using Markov inequality $\uparrow$ # of requests per sec

$$\leq \frac{E[X]}{1000} = \frac{1}{20}$$

$\rightarrow$ We can get a tighter bound on the probability if we also know the variance of the RV.

Let $Y = (X - \mu)^2$. Then $E[Y] = \sigma^2$.

$\uparrow$ mean of X $\uparrow$ Variance of X

We can apply the Markov inequality to Y .

This will give

$$P(Y \geq a^2) \leq \frac{E[Y]}{a^2}$$

$\uparrow$
 $(X-\mu)^2$

$$= \frac{\sigma^2}{a^2}$$

$$\Leftrightarrow P(X-\mu \geq a \text{ or } X-\mu \leq -a) \leq \frac{\sigma^2}{a^2}$$

$$\Leftrightarrow P(|X-\mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Chebyshev inequality

Example: Average height is 3.5 Ft, and
the variance is $\sigma^2 = \frac{1}{2}$ Ft.

What is the $P(\text{kid} \geq 9 \text{ Ft})$?

$$P(|X-3.5| \geq 5.5) \leq \frac{\frac{1}{2}}{(5.5)^2}$$

$= 0.008$ small!